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Professional Profile

Ninth-term Computer Science student with more than three years building software continuously since 2022 and over two years of professional experience. I am defined by my adaptability and my agency: I believe every problem has a solution. I am looking for a stable opportunity that lets me apply my technical experience and keep growing professionally.

Professional Experience

Independent Consultant Software & AI Consultant

Remote
July 2026 – Present

- Designing and building a RAG system for a Peruvian land-surveying firm, defining requirements and scope directly with the client.

CoFoundy - Software Consultancy Software Engineer

Remote
Mar 2025 – May 2026

- Led the design and implementation of the core backend of multiple production products, making the key technical decisions on architecture, stack and applied patterns.
- Designed modular architectures with NestJS applying Domain Driven Design, including business logic, authentication with JWT and OAuth2, persistence with PostgreSQL and caching with Redis.
- Implemented real-time communication with WebSockets, integrating JWT-based authentication and validating behavior under load through stress testing.
- Developed features in Next.js leveraging Server Actions, unifying backend logic inside the frontend where the case allowed.
- Integrated and configured low-code solutions such as Moodle, owning deployment and configuration in production environments.
- Containerized every service with Docker and defined CI/CD pipelines (GitHub Actions + Railway) to automate delivery and keep production stable.
- Broke requirements down into features and actionable tasks for the team, prioritizing delivery and providing technical follow-up.

SkyTech Software Developer

Remote
Jan 2024 – Feb 2025

- Implemented backend services with MQTT for a vehicle fleet management system used by real customers.
- Developed frontend features with Next.js within the product team, executing assigned tasks and taking part in code reviews.
- Picked up team work dynamics under agile methodologies: task estimation, iterative delivery and technical communication with the team.

Education

Universidad Nacional de Ingeniería
Computer Science - 8th term, Top Decile [\[certificate\]](#) [\[transcript\]](#)

Lima, Peru
April 2022 – Present

- Admitted in 2021 through the National School Admission exam (IEN), from my last year of high school at age 16, preparing solely with YouTube and books. This reflects the discipline and effort that define me.
- **Relevant coursework:** Computer Infrastructure (AWS labs with VPC, EC2, Lambda, RDS, S3, IAM, EBS and CloudWatch); Software Development (DevOps, Infrastructure as Code with Terraform and Kubernetes, containerization with Docker, Git flow, testing); Artificial Intelligence; Concurrent and Distributed Programming; Analysis and Design of Algorithms; Data Structures.

Universidade Estadual de Campinas (UNICAMP) - Academic Exchange

São Paulo, Brazil

Institute of Computing [\[acceptance letter\]](#)

July 2025 – Nov 2025

- During this exchange period I interacted with Brazilian professors and undergraduate and graduate students, which broadened my perspective and knowledge, as well as let me work with interdisciplinary people.
- I interacted with members of the Computer Networks Laboratory (LRC — Laboratório de Redes de Computadores), taking part in their technical discussions and research environment.
- **Courses taken:** Distributed Systems; Internet of Things; Software Verification, Validation and Testing; Bioinformatics.

Winter Camp in Artificial Intelligence — IA-PUCP

Lima, Peru

Pontifical Catholic University of Peru [\[evidence\]](#)

August 2026

- Selected as one of 41 participants from more than 100 applicants for the PUCP Artificial Intelligence Group intensive programme, covering Deep Learning, Natural Language Processing, Computer Vision, Recommender Systems and Explainable AI.
- Developing a group model to predict mass movements from satellite imagery and climate data.

Projects

AWS Cloud Infrastructure Labs

2026-I

AWS VPC, EC2, Lambda, RDS, S3, IAM, EBS, CloudWatch

Hands-on academic work completed for the Computer Infrastructure course at Universidad Nacional de Ingeniería, focused on secure, highly available and event-driven architectures.

- Configured VPC labs with public and private subnets across multiple Availability Zones, an Internet Gateway, route tables and least-privilege IAM roles.
- Designed, as a course assessment, an event-driven architecture where uploads to a private S3 bucket invoke Lambda to persist metadata in RDS, with EC2 access, additional EBS storage and CloudWatch alarms.
- Justified design decisions involving RDS isolation, EBS persistence, decoupling for high-volume workloads and connectivity for Lambda functions inside a VPC.

IoT System with Azure Cloud Services

Oct 2025 – Nov 2025

Azure IoT Hub, Azure Functions, MQTT, WebSockets, Next.js

Project built during the academic exchange at UNICAMP for the Internet of Things course, together with two graduate friends, where I specifically owned the software and the architecture. They, in turn, handled the physical and electronics side.

- Designed a scalable IoT architecture using Azure IoT Hub as an MQTT broker, supporting multiple devices and clients through a topic hierarchy.
- Developed serverless functions for real-time data processing and storage.
- Implemented a Next.js dashboard with its own domain and real-time visualization.

Distributed Log System with Consensus Algorithms

Oct 2025 – Nov 2025 Python, GCP, Lamport, Bully

Project built during the exchange for the Distributed Systems course, working with Google Cloud Platform, implementing and evaluating consensus algorithms such as Lamport and Bully, with response-time and throughput metrics.

- Implemented Lamport logical clocks for event ordering and the Bully algorithm for leader election.
- Deployed the system across multiple virtual machines and evaluated its behavior in different geographic zones.

TutorAI

May 2025 – Jul 2025

FastAPI, ElevenLabs

Built for the Human-Computer Interaction course in the seventh term: an AI-powered tutor for English learning, developed with FastAPI and ElevenLabs.

- Developed an adaptive AI assistant for English learning, with content generation and personalized feedback through voice APIs.
- Designed and ran usability tests with real users: we recorded university students' usage sessions, analyzed the recordings and extracted actionable conclusions to iterate on the product.

Multi-Agent System with CRDTs [\[certificate\]](#)

Sep 2024 – Nov 2024

FastAPI, LangGraph, Next.js, Redis, Yjs

Project presented at UNI's ABET fair, where it was exhibited and competed against other works in the program. We designed a real-time collaborative experience powered by intelligent agents.

- Designed the architecture for real-time collaborative editing using CRDTs (Yjs over Redis).
- Implemented the intelligent agent systems with LangGraph, exposed via FastAPI.
- Took part in the Next.js frontend development, integrating collaborative editing and the agent workflows.

Others

2022 – 2024

- **AI Forms Platform** (Feb 2024) — FastAPI, LangGraph: Universidad Continental hackathon. Backend with authentication and roles, and integration with LangGraph to generate dynamic forms from natural-language requirements.
- **Bluetooth Vehicle Control** (Nov 2023) [\[certificate\]](#) — ESP32, Bluetooth, Wi-Fi, HTTP, JS: Project exhibited at the Expociencia fair. I designed the client → backend → local client → ESP microcontroller architecture, connected over Bluetooth to the vehicle, explaining the lifecycle of an HTTP request live.
- **GeoBus Perú** (Jul – Dec 2023) — Node.js, Express, MongoDB, MQTT: Backend for simulation and control of vehicle fleets (Lima's Metropolitano corridors and UNI's internal transport).
- **Curriculum Map Visualizer** (Mar – Jun 2023) — JavaScript, vis-network, localStorage: Generic client-side web app for any UNI curriculum map, with persistence of academic progress.
- **Geographic Map of COVID-19 in Peru** (Sep – Dec 2022) — Python, Matplotlib, web scraping: Scraping of open data from the Ministry of Health and generation of monthly maps by department, browsable from a static HTML interface.

Hackathons & Awards

NASA Space Apps Challenge Campinas 2025 [\[certificate\]](#)

Nov 2025, Semifinalist

During the academic exchange in Brazil, I led an international team to simulate a lunar route using satellite imagery and NASA open APIs.

- Led an international team (Brazil, Bolivia, Peru), competing against teams from across the region.
- Implemented concurrent processing of geospatial imagery and real-time web visualization, integrating NASA public APIs.

International Potato Center Hackathon [\[certificate\]](#)

Apr 2024 – Feb 2025, World Finalist

Competing against international teams for 11 months, we presented our solution to a panel that included representatives from Amazon Web Services, Microsoft and research groups, reaching the world final of the challenge.

- We designed a hybrid deep-learning model with time series to predict potato crop characteristics based on the environment.

- I owned the data engineering (training dataset preparation and cleaning), the model implementation, its exposure via API and the web page development.

IMCA Hackathon [\[certificate\]](#)

Dec 2024, 1st place

Optimization of agricultural irrigation through Physics-Informed Neural Networks (PINNs), in a multidisciplinary team with a mathematician holding a master's degree, members of the agro-industrial sector and students from UNALM and UPCH.

- Design of the PINN-based solution.
- Joint work with domain experts from the agricultural sector.

Leadership & Activities

ACECOM - Scientific Association Specialized in Computing

Lima, Peru

Active member of the web development area since January 2024, where we combine technical training, mentorship and public project showcases. Usual stack of my contributions: Remix.

- Taught a web development workshop in April 2026 for faculty students (3rd to 6th term), teaching backend, secure backend and frontend concepts, as well as deployment with Railway and Git fundamentals; I answered questions live throughout the session.
- I take part on a recurring basis in the association's project fairs, presenting the area's developments.
- Automated internal processes: Discord integrations, cron jobs and a spam detector for email.

Volunteer academic tutoring — UNI

Lima, Peru

- Volunteer tutor for the Network-Centric Computing course (Sep 2024 – Dec 2024) [\[certificate\]](#), supporting academically at-risk students on topics such as the OSI model, security, cloud service models (IaaS, PaaS, SaaS) and cloud types (public, private, hybrid).
- Tutor for the Software Development course from Mar 2026 – Present [\[certificate\]](#), supporting at-risk students and the tutoring team on DevOps, IaC and software engineering practices.

Skills

Languages: TypeScript, Python, Java, Go (adaptable to any language). **Backend:** NestJS, FastAPI, Spring Boot, Express. **Frontend:** React, Next.js, Remix, Astro.

Cloud: AWS (VPC, EC2, Lambda, RDS, S3, API Gateway, IAM, EBS, CloudWatch), Azure, GCP. **DevOps & IaC:** Docker, Terraform, Kubernetes, GitHub Actions, CI/CD. **Databases:** PostgreSQL, MongoDB, Redis. **AI:** TensorFlow, LangGraph, Claude Code, Codex.

Spoken Languages: Spanish (Native); English (Advanced); Portuguese (Advanced).

Currently deepening Amazon Web Services and applied Artificial Intelligence. Working environment: Arch Linux with Hyprland.